

# THE ZERO PARADOX

## ZP-E: Bridge Document

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This document is the cross-framework synthesis layer of the Zero Paradox. It imports from ZP-A (lattice algebra), ZP-B (p-adic topology), ZP-C (information theory), and ZP-D (Hilbert space state layer). It provides three formal inserts: DA-1 (Instantiation as Execution), DA-2 (Instantiation Succession), and DA-3 (Perspective-Relative Cardinality). AX-1 (Binary Snap Causality) is retired. Its content was split in two: the shape of the snap is proved, as Theorem T-SNAP, and that the snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2). `tsnap_holds_but_nothing_moves` shows T-SNAP does not carry it. With DA-2, the directed instantiation tree is formally licensed. With DA-3, cardinality is shown to be position-dependent within the instantiation structure.

Illustrated Companion: A paired ZP-E Illustrated Companion document provides accessible explanations and visual summaries of the bridge derivations in this document. Readers new to the framework are encouraged to start with the companion.

A note on framework scope. The four layers (ZP-A through ZP-D) are not claimed to be the only mathematical domains in which the Zero Paradox structure appears. They are chosen because they cover the canonical mathematical languages for describing state: algebraic order structure (ZP-A), topology and metric geometry (ZP-B), information and complexity theory (ZP-C), and functional/Hilbert space analysis (ZP-D). The claim is that the same structural limit appears across each of the four frameworks addressed here — not that these four are the only possible witnesses.

Across all four layers, the same structural limit appears at  $\perp$  — but each framework names it differently: ZP-A identifies  $\perp$  as the global minimum, below which the lattice's ordering relation cannot descend; ZP-B finds the p-adic valuation undefined (or  $+\infty$ ) at 0, where clopen separation is total — 0 and every nonzero element lie in disjoint clopen classes; ZP-C finds surprisal unbounded above at  $\perp$ , so no finite external description can contain  $\perp$ ; ZP-D maps 0 to a basis vector orthogonal to every nonzero state, from which no return is possible. DA-1 establishes these as descriptions of the same structural limit across four independent mathematical languages.

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### Formal Insert DA-1: Derived Proposition — Instantiation as Execution

DA-1 is a Derived Proposition: instantiation as execution, the first Lean formalization of DA-1 (axiom-free, conditional on DP-2, `Snap.lean §VI`), with incompressibility read as self-description (ZP-C D1 + AIT) and CC-2/R3 grounding.

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#### I. The Gap DA-1 Closes

The T-BUF chain from ZP-C established three results:

- L-RUN: The transition  $c_0 \rightarrow c_1$  is a nonzero state transition. (ZP-C — Derived)
- TQ-IH: No program outputs  $\perp$  without a nonzero intermediate configuration state. (ZP-C — Derived by L-RUN)
- T-BUF: At  $P_0$ , execution is structurally guaranteed; that execution state is  $\epsilon_0$  in the semilattice. (ZP-C — Candidate Theorem pending DA-1)

T-BUF was labelled Candidate because Step 2 asserts that a configuration at  $P_0$  is a live machine state — that instantiation at  $P_0$  constitutes an execution event, not a static description. ZP-C L-INF supplies one mathematical premise:  $\perp$  at  $P_0$  has unbounded surprisal — no finite external interpreter can hold it as a static description. ZP-A CC-2 supplies a second, structural basis:  $\perp = \{\perp\}$  is a self-containing object with no external interpreter by structure. DA-1 (§ IV below) provides the derived proposition that closes T-BUF Step 2.

## II. The Two Senses of a Configuration at $P_0$

Sense A — Descriptive:  $x$  exists as a string — a finite syntactic object that has been written down or specified. The machine it describes has not necessarily been instantiated.  $P_0$  is a property of the string. The string is inert.

Sense B — Instantiated:  $x$  exists as the current configuration of a running machine. The machine is executing.  $P_0$  is a property of the live configuration. The configuration is active.

## III. Design Principle DP-2 — Execution Distinguishability

The two senses of § II share the same output value ( $\perp$ ) but differ in machine state. This separation — output value is not machine state — is made precise by Design Principle DP-2.

### Design Principle DP-2 — Execution Distinguishability

Machine states carry execution history independently of output values. A machine in state  $c_1$  can output  $\perp$  (the null value) while being in a configuration entirely distinct from a machine in state  $c_0$ . The post-execution null and the pre-execution null are different instances — same output value, different machine state.

Lean formalization (Snap.lean §VI): TrackedOutput separates value : MachinePhase from state : MachinePhase.  $\text{preInstantiation} = \langle c_0, c_0 \rangle$ ;  $\text{postInstantiation} = \langle c_0, c_1 \rangle$ . Theorem `dp2_execution_distinguishability` proves:  $\text{preInstantiation.value} = \text{postInstantiation.value}$  (same output)  $\wedge$   $\text{preInstantiation.state} \neq \text{postInstantiation.state}$  (distinct machine states). Proved axiom-free.

Given DP-2, DA-1 is Lean-formalizable at the minimal-path level. Theorem `da1_minimal_path` (Snap.lean §VI) establishes four conjuncts: (1) before and after instantiation, the output value is the same ( $\perp$ ); (2) the machine states are distinct ( $c_0 \neq c_1$ ); (3) the machine was at  $c_0$  before; (4) the machine is at  $c_1$  after. Proved axiom-free — no Kolmogorov complexity, no ZF+AFA required. The "return to null" after instantiation is  $\text{postInstantiation}$  (output  $\perp$ , state  $c_1$ ) — not  $\text{preInstantiation}$  (output  $\perp$ , state  $c_0$ ). Irreversibility (ZPB C3, ZPE `t_snap_irreversible`) blocks any path back to  $\text{preInstantiation}$ .

## IV. Proposition DA-1

### Proposition DA-1 — Instantiation as Execution

Claim: The instantiation of a machine configuration  $c_1$  at the incompressibility threshold  $P_0$  is an execution event in the sense of L-RUN. It is not a static description of a machine. It is a machine in state  $c_1$ .

Formal core (Lean): DP-2 (§III) — TrackedOutput separates output value from machine state. `da1_minimal_path` proves: one act of instantiation moves  $c_0$  to  $c_1$  regardless of output value. The returned  $\perp$  occupies the bottom role again; reading it as a NEW null rather than the prior  $c_0$  is C-DA2, a modelling commitment. The axiom-free result is the role, not the novelty.

Structural grounding: ZP-A CC-2 ( $\perp = \{\perp\}$ ) establishes  $\perp$  as a Quine atom under ZF + AFA — a set that is its own singleton, admitting no external interpreter. ZP-A R3 derives the immediate consequence: a self-containing object cannot be a static description awaiting external instantiation. ZP-C L-INF provides independent informational corroboration:  $\perp$  has unbounded surprisal, exceeding the capacity of any finite interpreter.

The three paths below are not parallel proofs of DA-1. Each argues independently for why the precondition DP-2 formalizes holds — why instantiation of  $\perp$  at  $P_0$  necessarily constitutes a first instruction fetch rather than a static description. DP-2 + `da1_minimal_path` derive  $c_0 \rightarrow c_1$  axiom-free once that precondition is established. The paths are convergent corroboration of the precondition; the formal derivation is DP-2's.

Path 1 — Structural (Self-Execution, ZP-J T-EXEC): Nothing exists outside the null space.  $\perp$  is prior to all differentiation — there is no external state, no prior cause, no position from which something else could execute  $\perp$ . If  $\perp$  executes at all, the only possible executor is  $\perp$  itself. A thing that executes itself contains itself, and the framework reads this as its own sole member:  $\perp = \{\perp\}$ , so  $\perp \in \perp$ . The argument is that this is forced by the impossibility of external execution, which makes it more than a free choice and less than a theorem. ZP-A R3 states the structural consequence: a self-containing object admits no external interpreter position. ZP-J T-EXEC proves that whatever fills the Quine-atom role is  $\perp$ :  $\text{IsQuineAtom}(q) \leftrightarrow q = \perp$ , proved axiom-free in Lean 4 (`ZeroParadox.t_exec_iff`). AFA (ZF + AFA) is the consistent set-theoretic home for this structure — chosen because the framework requires it, not the reverse. ZP-A CC-2 ( $\perp = \{\perp\}$ ) is a Forced Metatheoretic Commitment: on a lattice, T-EXEC proves that whatever fills the Quine-atom role is  $\perp$  (axiom-free, given the field `bot_self_mem`); in AFA set theory  $Q = \{Q\}$  is a theorem (Aczel 1988, Example 1.3), and that the framework's  $\perp$  is that set is the argued metatheoretic step (see Remark R-AFA).

Path 2 — Informational (ZP-C L-INF): Independently, the surprisal  $I(n) = n$  at ball-hierarchy depth  $n$  is unbounded — for any finite  $M$ ,  $\exists$  depth  $n$  with  $I(n) > M$ .  $\perp$  corresponds to the limit point  $0 \in Q_2$ ; its informational content exceeds every finite bound. Any finite external interpreter can hold only a finite informational bound;  $\perp$  exceeds every such bound. Note: Path 2 is motivational context, not a formal path to the conclusion. The step "exceeds every finite bound  $\rightarrow$  therefore necessarily executing" is a foundational commitment, not a derivable claim. It asks what it means for a mathematical structure to instantiate rather than merely satisfy conditions — a question no computability library answers. Forward paths: (a) a new axiom explicitly committing to this bridge; (b) a connection to Chalmers' notion of implementation; (c) the dissolution argument in The Philosophical Question That Started This — the separability of description and instantiation is the assumption the framework dissolves, not a gap it must close from the outside.

Path 3 — Formal bridge: Incompressibility as Self-Description (ZP-C D1 + standard AIT): The preceding paths establish that  $\perp$  admits no external interpreter. This path provides the formal bridge from that negative claim to the positive claim (necessarily executing). In the standard Turing model (D7), a machine configuration  $x$  exists in one of two states: (A) Static description —  $x$  exists as a string specified but not yet

being executed; some external program  $p$  ( $|p| < |x|$ ) generates  $x$  when run, so  $x$  is a description awaiting a separate execution event by an external generator. (B) Live execution —  $x$  is the current configuration of a running machine. These are exhaustive in the Turing model: either  $x$  has a shorter external generator, or it does not. At  $P_0$ , ZP-C D1 gives  $K(c_1|n)/|c_1| = 1$ :  $c_1$  is algorithmically incompressible. No external program  $p$  exists with  $|p| < |c_1|$  such that  $U(p, n) = c_1$ . State (A) requires such a  $p$  — and no such  $p$  exists at  $P_0$ . State (A) is therefore eliminated by the Kolmogorov condition. Since (A) and (B) are exhaustive and (A) is eliminated,  $c_1$  is in state (B): it is executing. Instantiation at  $P_0$  is not the placement of a description to be executed later — there is no shorter prior description to execute. Instantiation and execution are the same act.

The formal grounding (DP-2, §III) and the three paths above operate at distinct levels, not as alternatives to one another. DP-2 + `da1_minimal_path` establish a conditional at the level of machine-state representation: if instantiation of  $\perp$  constitutes a first instruction fetch in the sense of D7, then  $c_0 \rightarrow c_1$  follows necessarily — axiom-free, proved by construction. DP-2 is grounded in D7 itself (the standard computational distinction between before-first-instruction and after-first-instruction states), which is prior to and independent of DA-1. The three paths above operate one level down: they argue for why the precondition holds — why instantiation of  $\perp$  necessarily constitutes a first instruction fetch at all. Path 1 (structural) argues that nothing external to  $\perp$  can execute  $\perp$  — therefore  $\perp$  must execute itself, which argues for  $\perp = \{\perp\}$  as more than a free choice and less than a theorem (ZP-A CC-2, a Forced Metatheoretic Commitment); ZP-J T-EXEC proves, axiom-free, that whatever fills the Quine-atom role is  $\perp$ . Path 2 (informational) provides motivational context — unbounded surprisal as a pointer toward why static holding is incoherent — but the bridge from unbounded surprisal (L-INF) to execution is a foundational commitment, not a derived claim. Path 3 (AIT) argues that incompressibility eliminates the static-description alternative. Path 1 is witnessed by `da1_closed_concrete` (ZP-K), which proves `IsQuineAtom( $\perp$  : MachinePhase)` and nothing computational; Path 3's witness is the `machinePhaseKleene` instance's `botCode_is_quine` field, a `KleeneStructure` requirement, not a second independent proof. The two are carried together by `da1_paths_unified` as a conjunction of witnesses; that they name one structural fact is the framework's reading. Path 2 identifies a missing principle; its forward resolution is in *The Philosophical Question That Started This*. Path 2 is context. (ZP-M R-M.1 provides a retrospective structural analysis of why the gap resisted formalization.)

Status: DERIVED PROPOSITION — primary formal grounding: DP-2 (§III, `TrackedOutput` construction). `da1_minimal_path` proved axiom-free in Lean (Snap.lean §VI): instantiation moves  $c_0$  to  $c_1$  regardless of output value. ✓ Path 1 (structural, ZP-J T-EXEC + ZP-K): witnessed by `da1_closed_concrete` : `IsQuineAtom( $\perp$  : MachinePhase)`, proved in `Kleene.lean`, which proves nothing computational. (Under `MachinePhase`'s `selfMem x := x =  $\perp$` , this reduces to  $(\perp = \perp) \wedge (\forall x, x = \perp \Rightarrow x = \perp)$  — structural closure enforced by typeclass design, not a set-theoretic derivation from ZF+AFA. See R-K.0.) Path 2 (informational, L-INF): FOUNDATIONAL COMMITMENT — a missing principle, not a missing proof. Forward: *The Philosophical Question That Started This*; ZP-M R-M.1 (retrospective structural analysis). Path 3 (computational, ZP-K Kleene): its witness is the `machinePhaseKleene` instance's `botCode_is_quine` field — a `KleeneStructure` requirement, not a second independent proof; `da1_paths_unified` carries it with Path 1 as a conjunction of witnesses. CC-1 ( $S_0 = \perp$ ): restated — ZP-J `cc1_derived` with `t_exec_iff` makes a Quine-atom start and a  $\perp$  start the same condition (axiom-free, Lean) — and not forced: on a carrier with a second point a valid sequence starts elsewhere (`ZeroParadox/Settheory/OntBridge.lean`). CC-2 ( $\perp = \{\perp\}$ ): ZP-J `t_exec_iff` proves  $\perp$  is the only occupant of the Quine-atom role (axiom-free); that  $\perp$  is the AFA set  $Q = \{Q\}$  is an argued metatheoretic commitment (see R-AFA). DP-2 (§III) — explicit. T-SNAP's premises, in its Lean form and in its prose argument, are stated under Premises of T-SNAP (DA-1 insert, § V). AIT (Kolmogorov complexity) outside Lean scope; Kleene fixed-point is the in-scope formal counterpart.

## V. Theorem T-SNAP — Binary Snap Causality

### Theorem T-SNAP — Binary Snap Causality

“Causality” in this name refers to the shape of the step, not to its occurrence.

Statement: The Binary Snap  $\perp \rightarrow \varepsilon_0$  is a derived consequence of  $P_0$ , L-RUN, TQ-IH, DA-1, and ZP-A D2, among the results the seven steps below cite. It is not an axiom. What it rests on differs between its Lean form and its prose argument: see Premises of T-SNAP, directly below.

Premises of T-SNAP, in two readings. Both readings are true, and neither replaces the other. In Lean, two of the premises, CC-1 and occurrence at the first step, are the hypotheses of `t_snap_given`, named in (i).

- (i) Lean form: the shape. `t_snap_derived` :  $c_0 \neq c_1 \wedge c_1 \neq c_0 \wedge \text{join } c_0 \ c_1 = c_1$  (ZeroParadox/Order/Snap.lean) takes no hypotheses, and `#print axioms` reports no axioms. It is a fact about the fixed two-state type `MachinePhase`, with the initial phase  $c_0$  as its bottom element  $\perp$  and the running phase  $c_1$  above it: the two phases differ, and joining them gives  $c_1$ . Nothing is passed to it as an argument; binary existence is built into the choice of that type, and the start at  $\perp$  into the statement’s choice of  $c_0$ . It fixes the SHAPE of the transition, not that one occurs: `tsnap_holds_but_nothing_moves` proves the same statement in a dynamics where every phase is a fixed point. Two of the premises as hypotheses: `t_snap_given` (same file) holds over any join-semilattice with bottom, taking  $S_0 = \perp$  (`hcc1`, CC-1) and  $S_1 \neq S_0$  (`hocc`, occurrence at the first step: the step at index 1 is taken) and giving the same shape at  $S_0, S_1$ . Its two inequalities restate `hocc`; only the join conjunct is derived, from A4. `t_snap_derived`’s statement is its `MachinePhase` instance at the sequence  $c_0, c_1, c_1, \dots$ , where `hcc1` holds by `rfl` and `hocc` reduces to  $c_1 \neq c_0$ . `MachinePhase` does not discharge occurrence: the sequence that stays at  $c_0$  is a state sequence in the same type and fails `hocc`. With no dynamics given, dropping either hypothesis makes the conclusion fail (examples beside it). `hocc` forces at least two points and does not force exactly two; AX-B1’s discreteness is not in that signature.
- (ii) Prose argument: the shape, then the occurrence. The seven steps below read that shape as a statement about the framework’s own state sequence. That reading uses commitments, among them AX-B1 (binary existence, ZP-B, the framework’s one substantive modelling commitment) at Step 4, and CC-1 ( $S_0 = \perp$ , a ZP-A Conditional Claim) for the start at  $\perp$  in Steps 4 and 6. That the transition OCCURS rests further on two things. The occurrence commitment: instantiation occurs (a machine configuration reaches  $P_0$ ). And Step 2, which imports DA-1: a configuration at  $P_0$  is executing. DA-1 is closed only given DP-2: § IV derives  $c_0 \rightarrow c_1$  from DP-2 once DP-2’s precondition is established, and calls its three paths convergent corroboration of that precondition; Path 1 argues through ZP-A CC-2, itself a commitment. Together they give that the Snap occurs, which is not a theorem. Given CC-1, its first-step form is the hypothesis `hocc` of `t_snap_given` (a sequence that first moves at a later index fails `hocc`). ZP-C L-INF supplies unbounded surprisal, and the step from there to forced execution is a foundational commitment (§ IV), not a mathematical consequence of L-INF alone.

Proof:

Note on Lean scope: the formal Lean proof of T-SNAP is a single term — `(l_run, tq_ih, rfl)` — `l_run` proves  $c_0 \neq c_1$  by `decide`, `tq_ih` is its symmetric form, and `rfl` closes the join in the `MachinePhase` semilattice (`t_snap_given` obtains that conjunct from A4, `bot_join`). What follows is the informal motivation for the modelling choices that let that term be read as a statement about the framework’s own state sequence.

- Step 1 —  $P_0$  identifies the incompressibility threshold. When  $K(x|n)/n = 1$ , the configuration string  $x$  is algorithmically random. (ZP-C D1)
- Step 2 — A configuration at  $P_0$  is necessarily executing: At  $P_0$ ,  $K(c_1|n)/|c_1| = 1$  (ZP-C D1) —  $c_1$  is incompressible, its own minimal program. No shorter external generator exists; the static description state is eliminated;  $c_1$  is in live execution (DA-1 Path 3 — from ZP-C D1 + AIT). Corroboration:  $\perp = \{\perp\}$  (ZP-A CC-2/R3); unbounded surprisal (ZP-C L-INF). (DA-1 § IV — Derived Proposition)
- Step 3 — Any instantiated execution passes through  $c_1$ . (ZP-C D7 — definitional;  $c_1$  is the first running configuration)
- Step 4 —  $c_1 \neq \perp$ . (ZP-C L-RUN — Derived;  $c_1$  has gained execution context not present in  $c_0 = \perp$ ; by AX-B1 this is a distinct, nonzero state)
- Step 5 — No program that executes produces only  $\perp$  configuration states. (ZP-C TQ-IH — Derived; execution trace  $\tau(p)$  contains  $c_1$  for any executing program  $p$ )
- Step 6 — In  $(L, \vee, \perp)$ ,  $c_1$  is an element strictly above  $\perp$ . By ZP-A D2, the transition  $\perp \rightarrow c_1$  is a valid state transition:  $c_1 = \perp \vee \epsilon_0$  for some  $\epsilon_0 \in L$  with  $\epsilon_0 > \perp$ . This transition is the Binary Snap.
- Step 7 — The transition is irreversible: algebraically by ZP-A R1 (no subtraction operator); topologically by ZP-B C3 (no continuous return path to 0 in  $Q_2$ ). These two grounds are sufficient. Conceptual correspondence: ZP-G AX-G2 ( $\text{hom}(X, 0) = \emptyset$  for  $X \neq 0$ ) expresses the same irreversibility in categorical language — ZP-G is downstream of ZP-E and is not a formal dependency of this proof.

Conclusion: The Binary Snap is a derived consequence, given the commitments under Premises of T-SNAP. AX-1 (Binary Snap Causality) is retired. Its content was split in two: the shape of the snap is proved, as Theorem T-SNAP, and that the snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2). `tsnap_holds_but_nothing_moves` shows T-SNAP does not carry it. ✓

Status: DERIVED — Cross-Framework. Premises: in two readings, under Premises of T-SNAP above; the Lean form fixes the shape with no hypotheses, `t_snap_given` takes CC-1 and occurrence at the first step as hypotheses, and the prose argument's shape and its occurrence rest on commitments, among them those named there. Dependencies cited in the steps: ZP-C D1, D7, L-RUN, TQ-IH; ZP-B C3; ZP-A D2, R1; ZP-E DA-1; ZP-J T-EXEC. CC-1 ( $S_0 = \perp$ ): restated — a Quine-atom start and a  $\perp$  start are one condition (ZP-J `cc1_derived`, `t_exec_iff`; axiom-free) — and not forced, since on a carrier with a second point a valid sequence starts elsewhere. CC-2 ( $\perp = \{\perp\}$ ): ZP-J `t_exec_iff` proves  $\perp$  is the only occupant of the Quine-atom role (axiom-free); that  $\perp$  is the AFA set  $Q = \{Q\}$  is argued metatheoretically (see R-AFA). Both remain commitments: CC-1 is a Conditional Claim, expressed through the Quine-atom role and not removed by it; CC-2 is a Forced Metatheoretic Commitment. Conceptual correspondence only: ZP-G AX-G2 (downstream of ZP-E; not a formal dependency).

#### Remark R- $\epsilon_0$ — On the Symbol Choice for the Minimum Snap Displacement

Note: this remark draws an informal structural analogy. No formal embedding of ZP's  $\epsilon_0$  into the Cantor-Gentzen ordinal is claimed or established here. The symbol  $\epsilon_0$  in Step 6 denotes the minimum element of  $L$  strictly above  $\perp$  — the least witness for the Binary Snap displacement. This symbol is chosen deliberately to coincide with the Cantor-Gentzen proof-theoretic ordinal.

### Remark R- $\epsilon_0$ — On the Symbol Choice for the Minimum Snap Displacement

The Cantor-Gentzen ordinal  $\epsilon_0$ . In ordinal arithmetic,  $\epsilon_0$  is the smallest fixed point of the map  $\alpha \rightarrow \omega^\alpha$ : equivalently,  $\epsilon_0 = \sup\{\omega, \omega^\omega, \omega^{\omega^\omega}, \dots\}$ . No finite  $\omega$ -tower reaches it — it is the minimum ordinal that cannot be generated from 0 by any finite iteration of the base operation. Gentzen established that transfinite induction up to  $\epsilon_0$  is necessary and sufficient to prove Con(PA). By Gödel's incompleteness theorem, PA cannot prove this from within. The ordinal  $\epsilon_0$  is therefore the minimum threshold at which finite arithmetic exhausts its own generative capacity.

The structural analogy. ZP's  $\epsilon_0$  occupies an analogous position in the state lattice. At  $P_0$ ,  $c_1$  satisfies  $K(c_1|n)/|c_1| = 1$  (ZP-C D1): it is algorithmically incompressible — no finite external program shorter than  $c_1$  generates it. Just as the Cantor  $\epsilon_0$  cannot be reached from 0 by any finite  $\omega$ -tower, ZP's  $\epsilon_0$  cannot be reached from  $\perp$  by any finite external description. Both name a structurally analogous object: the minimum witness for a transition that exhausts the finite generative hierarchy below it.

The proof structures are parallel. Gentzen locates the minimum ordinal strength at which PA cannot describe its own consistency from within. ZP locates the minimum state displacement at which no external program can hold  $\perp$  as a static string — where unbounded surprisal (ZP-C L-INF) and self-containment (ZP-A CC-2) together eliminate any external interpreter position. In both cases the exhaustion of finite description is not a deficiency but a structural consequence: the system is necessarily executing at  $\epsilon_0$ .

What is not claimed. ZP does not assert that L is an ordinal structure, or that ZP's  $\epsilon_0$  is literally the Cantor ordinal under a formal embedding into the p-adic/lattice framework. The analogy is motivational: both  $\epsilon_0$ s mark the minimum witness for incompressibility relative to a finite base. A formal embedding — showing that the Cantor  $\epsilon_0$  is order-isomorphic to or embeds into the p-adic completion of L at  $\perp$  — remains an open question and would constitute a strengthening of this claim.

### Remark R-AFA — Why Foundation is Ruled Out; AFA as Forced Replacement

CC-2 — the identification of  $\perp$  with the Quine atom  $\perp = \{\perp\}$  — is a set-theoretic step beyond ZP-A's algebra: it is not derived from A1–A4. The choice of ZF + AFA over ZF + Foundation that it entails is nonetheless not arbitrary: it is forced by the membership structure the framework's bottom must have.

Foundation rules out the bottom (structural). The framework's bottom is the Quine atom  $\perp = \{\perp\}$ : it is a member of itself,  $\perp \in \perp$ . ZF + Foundation (the Axiom of Regularity) forbids exactly this — under a well-founded membership relation no set lies on a membership cycle of any length, so in particular no set is a member of itself. This is a theorem of ZF + Foundation, not a framework-specific stipulation; it is machine-checked here as `no_quine_atom / no_membership_cycle` (ZeroParadox/Settheory/Wall.lean, footprint [propext, Quot.sound], choice-free). A Foundation-respecting universe therefore cannot host  $\perp$  at all.

The framework's readings of  $\perp$  corroborate this (they do not prove it). R3 (ZP-A) —  $\perp = \{\perp\}$  has no describer position external to itself — and L-INF (ZP-C) — the 2-adic surprisal  $I(n) = n$  is unbounded, so no finite interpreter can hold  $\perp$  — each describe an object with no external vantage and unbounded self-referential depth. That is exactly the profile of a non-well-founded, self-membered set: the two readings motivate why  $\perp$  must be the circular object Foundation excludes, rather than serving as independent proofs of the exclusion.

AFA as the forced replacement. Under ZF + AFA, Quine atoms ( $x = \{x\}$ ) are the minimal non-well-founded objects: the membership structure of  $\perp = \{\perp\}$  is circular ( $\perp \in \perp \in \perp \in \dots$ ), so it lives in a non-well-founded metatheory, and AFA is the minimal such theory that supplies exactly one of them.

### Remark R-AFA — Why Foundation is Ruled Out; AFA as Forced Replacement

What is machine-checked (the requirements framework). The question of which host theory can admit  $\perp$  is reduced to three requirements a host must supply — Foundation-freeness, a Quine atom, and its uniqueness — and largely closed in the framework's set-theory layer (the QuineHost development, ZeroParadox/Settheory/QuineHost.lean, formalized later in the ZP-J AFA layer). Proved there: Foundation-freeness is forced for any such host (quineHost\_not\_wellFounded, axiom-free); ZF + Foundation is excluded (zfSet\_no\_quine\_bottom — no set is self-membered) and Boffa's anti-foundation axiom is excluded as too permissive, admitting a proper class of atoms rather than one (boffa\_fails\_unique); and the requirements are realizable — a concrete one-atom model exhibits them (oneAtom\_not\_wellFounded, axiom-free) and AFA is the canonical example (afaStructure\_isQuineHost). So the AFA choice is not merely argued: the forcing and the realizability are settled results.

What remains — a narrow Forced Metatheoretic Commitment. With the forcing and realizability proved, one step is not a theorem: that a Quine atom and its uniqueness are the right two requirements to demand — the commitment to  $\perp = \{\perp\}$  as the minimal option consistent with A4's additive-identity role (it has exactly one member, itself, and introduces no internal differentiation; any extension  $\perp = \{\perp, x\}$  with  $x \neq \perp$  would add members carrying their own membership chains). This is a Forced Metatheoretic Commitment — stronger than a free choice, weaker than a theorem — not a Conditional Claim; the set-membership face  $\perp \in \perp$  stays metatheoretic, while the occupant of the Quine-atom role is machine-checked to be  $\perp$  (t\_exec, ZeroParadox/Settheory/SetTheoryAFA.lean, axiom-free). The named falsifier is correspondingly narrow: a differently-motivated requirement set that hosts  $\perp$  without demanding a unique Quine atom would reopen this choice — it would not touch the forcing or the realizability, which are settled.

## VI. Effect of T-SNAP on Downstream Results

Remark R-DA1: AX-1 is retired, and its content was split in two. Results in ZP-E that previously depended on AX-1 as an axiom now depend on T-SNAP, a derived theorem, for the shape, and, wherever they use that the Snap happens, on the occurrence commitment (instantiation occurs) together with DA-1. T5 (Iterative Forcing Theorem) depended on AX-1 and is restated below. T4 (Unified Snap Description) carried AX-1 as an axiom label on the causality component — that label is upgraded to Derived — T-SNAP for the shape, with occurrence still a commitment. DA-1 is additionally upgraded from Design Principle to Derived Proposition: grounded in ZP-A CC-2 ( $\perp = \{\perp\}$ ) and R3, with ZP-C L-INF as independent corroboration. The intentional axioms of the system are now: AX-B1 (binary existence), AX-G1 (initial object), AX-G2 (source asymmetry). AX-1 is retired: its shape is Theorem T-SNAP, and the occurrence commitment is not on this list because it is a commitment, not a named axiom. Additional structural commitments are carried as typeclass fields: A4 (bot\_join, ZPSemilattice — standard semilattice algebra), AFAStructure.quine\_unique and bot\_self\_mem (the Quine-atom role), KleeneStructure.botCode\_is\_quine (computational closure). Like named axioms these are assumptions, carried as hypotheses by every theorem that uses them; #print axioms does not surface typeclass fields. Unlike named axioms they can be cheap to meet: every ZP-A lattice supplies AFAStructure trivially.

## T5 (Iterative Forcing Theorem) — restated

T5 (restated). Selection. In the ordinal chart the next rung is the snap re-seeded one step past the current one (succession\_succ), and none lies between two consecutive rungs (ZeroParadox/Ordinal/SnapSuccession.lean § I). Carried into the two-state lattice by a monotone map that sends the tower's stages to  $c_0$  and  $\varepsilon_0$  to  $c_1$ , no ordinal below  $\varepsilon_0$  is sent to  $c_1$ , and every fixed point of  $\alpha \mapsto \omega^\alpha$  is (snap\_unconditional, hfp\_from\_epsilon\_zero). The two faces of  $\varepsilon_0$  carry the two directions: as the supremum of the stages, every ordinal below  $\varepsilon_0$  lies under one of them, so nothing below fires (fundamentalSeq\_cofinal); as the least fixed point of  $\alpha \mapsto \omega^\alpha$  it lies below every other, so every landing from  $\varepsilon_0$  up fires (epsilon0\_min\_eq\_max). The placement at  $\varepsilon_0$  is a hypothesis there ( $h\varepsilon_0$ ), which the Lean names the alignment hypothesis; deriving it from the 2-adic structure is open, as the Classical.choice inversion conjecture (ZeroParadox/Ordinal/Incompleteness.lean). The rungs are the iterative bottoms; none of them is  $\perp$  (epsilon0\_ne\_bot; for every rung, Ordinal.epsilon\_pos). On an arbitrary lattice no step is selected: 0, 2, 4, ... on the natural numbers satisfies A1–A4 and AX-B1.

Iteration. No step is guaranteed. A state sequence moves only upward (T3) and never returns (R1, t\_snap\_irreversible). Where the next increment is already absorbed ( $S_n \vee \alpha_n = S_n$ ), the state stays the same, which T-SNAP permits (tsnap\_holds\_but\_nothing\_moves); different runs need not reach the same height. If from some step  $x$  on every new increment is already absorbed, then  $S_x$  is the supremum of the run: the value that run reaches in the limit. Short of a top, reaching it cannot be confirmed at any finite step; a run that reaches a top state ( $c_1$  on the two-state carrier) is known there to stay.

Reading:  $\varepsilon_0$  is both at once for the ordinal tower: the least fixed point of  $\alpha \mapsto \omega^\alpha$  and the supremum of its stages (epsilon0\_min\_eq\_max).

The handle T5 is kept. “Forcing” names the shape of each step taken, not that any step is taken.

## Formal Insert DA-2: Instantiation Succession — The Multiple- $\perp$ Result

Formally licenses the directed instantiation tree

### I. The Gap DA-2 Closes

Given the premises stated under Premises of T-SNAP (DA-1 insert, § V), DA-1 and T-SNAP give the Binary Snap on reaching  $P_0$  within an instantiation. Three questions remain open after DA-1:

- CC-1 (ZP-A) posits  $S_0 = \perp$  — not derivable from A1–A4 alone (ZP-J restates it as an equivalence with starting at a Quine atom, cc1\_derived and t\_exec\_iff, and does not force it). This leaves open whether  $\perp$  is unique across all instantiations or whether each instantiation carries its own  $\perp$ .
- ZP-B R1 distinguishes universal structure from universe-contingent parameters.  $\varepsilon_0$  is contingent per instantiation. Whether  $\perp$  is similarly contingent is not addressed in ZP-A through ZP-D.
- Occurrence fence. T-SNAP fixes the SHAPE of each step. It does not establish that any step is taken: tsnap\_holds\_but\_nothing\_moves exhibits a model in which T-SNAP holds and nothing moves. Every “fires” below narrates the commitment that instantiation occurs, not a consequence of the theorem.
- T-SNAP fires wherever  $P_0$  conditions are met. If the terminal state of instantiation  $I_n$  satisfies  $P_0$  conditions, T-SNAP should apply — but this requires formally connecting that terminal state to a  $\perp$  role-occupant, read as a new  $\perp$  under C-DA2. DA-2 provides this connection.

## II. Why ZP-B C3 is Not Violated

C3 prohibits a continuous path from  $x \neq 0$  back to the same 0 in  $Q_2$ . DA-2 does not require a return path. The irreversibility of C3 is preserved within each instantiation. What crosses the instantiation boundary is not a path in  $Q_2$  — it is the generation of a new  $Q_2$  with its own metric, its own  $\perp$ , its own  $\varepsilon_0$ . C3 quantifies only over paths within a single topological space and has nothing to say about the boundary between spaces.

More precisely: C3 rules out a continuous return to 0 WITHIN one  $Q_2$ . It does not follow that a recurrence of  $\perp$  must be a structurally distinct  $\perp$  — that step needs the separateness of the spaces, which is a modelling commitment (C-DA2) and not something the topology supplies. The topology does not enforce novelty; it is silent on it, and in the 2-adic chart the arc reapproaches the same 0 (snap\_arc\_z2\_loop).

## III. Definitional Alignment DA-2 — Instantiation Succession

| Definitional Alignment DA-2 — Instantiation Succession                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Claim: A state $S$ in instantiation $I_n$ satisfies the structural role of $\perp$ for instantiation $I_{n+1}$ if and only if it satisfies A4 relative to all subsequent joins in $I_{n+1}$ : |
| $S \vee x = x$ for all $x$ in the semilattice of $I_{n+1}$ .                                                                                                                                  |

Grounding: A4 is the load-bearing axiom of ZP-A — it defines  $\perp$  as the additive identity under  $\vee$ , the element that contributes nothing to any join and is therefore present in everything above it. DA-2 does not redefine  $\perp$ . It clarifies that the CC-1 identity condition ( $S_0 = \perp$ ) can be satisfied by any state meeting A4's algebraic conditions — not only by a cosmologically primitive  $\perp$ . The identity condition is structural, not historical: what matters is the algebraic role a state plays in the subsequent semilattice, not where it came from.

The terminal state of  $I_n$  arrives at  $I_{n+1}$  carrying the accumulated join of everything in  $I_n$ 's sequence. It is structurally  $\perp$  to  $I_{n+1}$  — contributing nothing to subsequent joins — while being informationally rich relative to  $I_n$ . This is the Zero Paradox instantiated at the inter-instantiation level: the terminal state is simultaneously a terminus and a foundation.

Status: DEFINITIONAL ALIGNMENT — no new axiom introduced. DA-2 is a clarification of the scope of CC-1 and A4. ✓

## IV. Conditional Claim C-DA2 — Novelty of Successive $\perp$

| Conditional Claim C-DA2 — Novelty of Successive $\perp$ (a modelling commitment)                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Commitment: each instantiation carries its OWN topological space, so the $\perp$ of $I_{n+1}$ is not an element of the $Q_2$ of $I_n$ . GIVEN that, no two instantiations share a $\perp$ , and succession is a chain (or tree) rather than a cycle. |
| Status: CONDITIONAL, not derived. The commitment above is the separateness of the spaces; it is assumed, not obtained from C3. Formerly labelled a Corollary and marked derived — retracted, see the note below. The handle C-DA2 is unchanged.      |
| What IS proved: that anything occupying the bottom role IS the bottom of its own lattice (t_iz_limit_is_new_null). That is an identity with the bottom already present, not the production of a second one.                                          |

Why this is NOT a proof, stated plainly because it was published as one. The step "the  $\perp$  of  $I_{n+1}$  is an element of a distinct topological space" is the CONCLUSION, not a consequence of C3. C3 quantifies over paths WITHIN a single space — it says no continuous path runs from any  $x \neq 0$  back to 0 inside one  $Q_2$  — and has nothing to say about whether the next instantiation's space is a different one. Assuming separateness and then deriving distinctness from it is circular. DA-2 likewise fixes identity conditions WITHIN an instantiation; it does not compare across two.

And the corpus measures against it where the question is concrete: in the 2-adic realization the arc returns to the SAME zero. `snap_arc_z2_loop` proves the tower starts at 0, that every finite stage is  $\neq 0$ , and that it converges back to 0; `tower_image_loops_to_seed` states the limit IS the seed value. So in that chart novelty does not merely lack a proof — it fails. C-DA2 stands as a modelling commitment about instantiations, and must not be cited as derived, nor supported by T-IZ or by `t_iz_limit_is_new_null`.

## V. The Directed Instantiation Tree

With DA-2 and C-DA2 in place, the global structure of instantiations is a forward-directed tree with no back edges:

- Each node in the tree is a  $\perp$  — the bottom element of one instantiation and the foundation of all successor instantiations branching from it.
- Each edge within an instantiation is a step in a monotone state sequence (ZP-A T3). Edges are irreversible (ZP-B C3).
- Branching at each node: every distinct outbound vector from the terminal state of  $I_n$  is a valid  $\epsilon_0$  for a distinct  $I_{n+1}$ . T-SNAP does not select among branches. Because  $\perp = \{\perp\}$  (ZP-A CC-2) is the single self-containing  $\perp$  with no internal differentiation, every  $\epsilon_0$  that represents a first differentiation in any direction is a valid outcome. The work here is done by the undifferentiated  $\perp$  of CC-2, not by T-SNAP: given that instantiation occurs, no direction is privileged, so the structure branches rather than threading a line. T-SNAP fixes the shape of each step; it does not supply the occurrence.
- No back edges: C-DA2 COMMITS to no instantiation reaching the  $\perp$  of any ancestor. This is the commitment, not a theorem - in the 2-adic chart the arc returns to the same 0 (`snap_arc_z2_loop`).

Remark R-DA2: T-SNAP fires wherever  $P_0$  conditions are met. DA-2 establishes that the terminal state of  $I_n$  satisfies those conditions for  $I_{n+1}$ . T-SNAP therefore applies across instantiation boundaries without modification. No new axiom is required. The multiverse is not the claim that T-SNAP fires in all directions simultaneously: it is the claim that  $\perp = \{\perp\}$  (ZP-A CC-2) has no internal differentiation and therefore no preferred direction for  $\epsilon_0$ . The multiverse of instantiations is the full set of all minimal differentiations available from the single self-containing  $\perp$  — a structural consequence of CC-2 + T-SNAP + DA-2 jointly.

## VI. The Zero Paradox Iterated

The paradox of  $\perp$  — simultaneously contributing nothing and being present in everything — propagates structurally at every branching node of the tree. Each node is:

- Nothing to its successor instantiations: it acts as additive identity under  $\vee$ , contributing nothing to any subsequent join.
- Everything to its successor instantiations: every state in  $I_{n+1}$  satisfies  $\perp_{n+1} \leq S$ , so the node underlies everything that follows.

The tree is the geometric shape of the Zero Paradox iterated across instantiations. The single-instantiation linear sequence was always a cross-section of a structure with this shape.

The complete picture: The Zero Paradox describes a forward-directed infinite tree where  $\perp_1 \rightarrow \dots \rightarrow S_{\text{terminal}}^1 \equiv \perp_2 \rightarrow \dots$ , where  $\equiv$  means structurally satisfies the role of, not is identical to. Each arrow within an instantiation is monotone and irreversible. Each  $\equiv$  crossing is not a path — it is a new instantiation of the universal structure.

## VII. Implications Within the Framework

**Structural Implication: Branching Tree Structure.** T-SNAP fixes the SHAPE of the Binary Snap — that if  $\perp$  transitions it goes to some  $\varepsilon_0 > \perp$ , and that the step is one-way. It does not establish that the transition is taken: that the Snap occurs follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2), and the NO-GO gauge `tsnap_holds_but_nothing_moves` exhibits a model in which T-SNAP holds and nothing moves. DA-2 establishes that any terminal state satisfying  $P_0$  conditions acts as  $\perp$  for a successor instantiation, generating a forward-directed branching tree. The branching comes from CC-2's undifferentiated  $\perp$  and the succession from DA-2, both conditional on instantiation occurring. Note: T-SNAP alone does not establish that it fires on all outbound vectors simultaneously — that universality is the scope of DA-2, not a direct consequence of the snap theorem itself.

**Monotonicity and Path Irrecoverability.** Within an instantiation, state sequences are monotone — no state can be decreased (ZP-A R1). Every state change is a join operation:  $S_n \vee \alpha$  for some increment  $\alpha$ . The algebra constrains only that the sequence be monotone, not which monotone path is taken. Each join adds informational content irreversibly. States are permanently encoded in the element's position in L.

**Ordinal Direction of State Sequences.** The monotone sequence (ZP-A T3) is a structural definition of directional asymmetry in state ordering. Irreversibility is C3 applied within an instantiation. The framework does not assume directional asymmetry — it derives it.

**Causal structure.** Every state is fully determined by the joins that produced it. The causal history of any state is encoded in its position in L. No effect without the join that produced it.

## Formal Insert DA-3: Perspective-Relative Cardinality

Cardinality as position-dependent measurement within the instantiation structure

### I. The Gap DA-3 Closes

DA-2 establishes that instantiations form a directed tree and that branching at each  $\perp$  node produces multiple successor instantiations. This raises a question about the cardinality of branching: is the fan at each node countably or uncountably infinite? The answer, which DA-3 formalises, is that this question is perspective-dependent — and DA-3 further conjectures (Claim DA-3-C1, deferred to OQ-E2) that this perspective-dependence is the same structural feature underlying the major cardinality anomalies of classical set theory.

### II. Perspective-Dependence of Branching Cardinality

From within instantiation  $I_n$ , an observer occupies exactly one branch of  $I_{n+1}$ . The other branches of  $I_{n+1}$  are not accessible via any path — C3 and monotonicity jointly prohibit it. From inside, the branching factor is always 1: the observer sees one branch. From outside — from a meta-level view of the tree — the

branching factor is the full fan of accessible outbound vectors.

Definition DA-3-D1 — Accessible Cardinality

The accessible cardinality of a position  $p$  in semilattice  $L$  is the cardinality of the set of states reachable from  $p$  by monotone sequences within the instantiation containing  $p$ .

The accessible cardinality from  $p$  is determined entirely by the structure of  $L$  above  $p$ . It is not an intrinsic property of a collection — it is a property of the relationship between a position and the states reachable from it. No position within any instantiation can access all cardinalities simultaneously. The meta-level view, which sees the full branching fan, is not a position any element of any instantiation can occupy.

Remark R-DA3-1: To observe the full branching fan, one would need to occupy a position outside all instantiations. That position would itself be a state in some semilattice, subject to the same rules. The meta-view is either another instantiation (in which case the tree has no privileged outside view) or  $\perp$  itself (in which case  $\perp$  is the only position from which the full structure is visible — the state that contributes nothing and is present in everything). The Zero Paradox’s name is more precise than it first appeared.

III. The Cardinality Hierarchy as Perspective-Relative

Cantor’s theorem establishes that for any set  $S$ ,  $|P(S)| > |S|$ , generating the hierarchy  $\aleph_0 < 2^{\aleph_0} < 2^{2^{\aleph_0}} < \dots$  DA-3 reframes this hierarchy not as a fixed ladder that mathematics climbs, but as a perspective-relative description of the branching structure of the instantiation tree, as seen from within different positions.

Claim DA-3-C1 — Perspective-Relative Absolute Cardinality

The appearance of absolute cardinality — cardinality as an intrinsic property independent of measuring position — is an artifact of treating the semilattice as having a view from outside. DA-2 and C-DA2 jointly prohibit such a view from within any instantiation.

The candidate claim (DA-3-C1) is that accessible cardinality from within any instantiation cannot replicate the view from outside. Whether specific independence results in classical set theory are formal expressions of this perspective-dependence is the conjecture that OQ-E2 is tasked with investigating.

Status: DEFINITIONAL ALIGNMENT + CANDIDATE CLAIM. DA-3-D1 and R-DA3-1 are definitional. DA-3-C1 is a candidate claim: structurally motivated within the framework; formal derivation of the connection between accessible cardinality and specific set-theoretic independence results is deferred to OQ-E2.

Updated Open Items Register

| Item                        | Status                                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AX-1: Binary Snap Causality | RETIRED — shape proved as T-SNAP; the snap occurs given the occurrence commitment and DA-1 | Its content was split in two: the shape of the Snap is proved, as Theorem T-SNAP (from L-RUN, TQ-IH and the bottom law, with no Lean kernel axioms), and that the Snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2). Premises under Premises of T-SNAP (DA-1 insert, § V); given CC-1, the first-step form of the Snap occurring is the hypothesis hocc of t_snap_given. |

| Item                                              | Status                                                                                                                                                                                                                                                                              | Description                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DA-1: Derived Proposition (DP-2 formal grounding) | CLOSED — DP-2 (formal core); CC-2 + L-INF + AIT (corroboration of precondition)                                                                                                                                                                                                     | Primary formal grounding: DP-2 (TrackedOutput, Snap.lean §VI) — <code>da1_minimal_path</code> proved axiom-free. Instantiation of $\perp$ moves machine from $c_0$ to $c_1$ ; output value is irrelevant to whether execution occurred. Informal corroboration: ZP-A CC-2 + R3 (structural); ZP-C L-INF (informational); AIT incompressibility (Path 3). |
| DA-2: Instantiation Succession                    | CLOSED — Definitional                                                                                                                                                                                                                                                               | Terminal state of $I_n$ satisfies A4 role of $\perp$ for $I_{n+1}$ . C-DA2 COMMITS to the novelty of each $\perp$ ; it does not establish it.                                                                                                                                                                                                            |
| DA-3: Perspective-Relative Cardinality            | CLOSED (definitional) / CANDIDATE (DA-3-C1)                                                                                                                                                                                                                                         | DA-3-D1 establishes accessible cardinality is position-dependent within the instantiation structure. DA-3-C1 (candidate): no position within an instantiation can replicate the outside view. Whether this connects formally to specific set-theoretic independence results is deferred to OQ-E2.                                                        |
| OQ-A1: Increment selection                        | OQ-A1b CLOSED by A1–A4; OQ-A1a: no reason to restrict to join-irreducibles (well-founded carriers); placement in the ordinal tower, for a monotone map sending the tower's stages to $c_0$ , conditional on the alignment hypothesis $h\varepsilon_0(\varepsilon_0$ sent to $c_1$ ) | T5 (Iterative Forcing Theorem), restated in DA-1 insert § VI. No step is guaranteed. On an arbitrary lattice no step is selected: 0, 2, 4, ... on the natural numbers satisfies A1–A4 and AX-B1.                                                                                                                                                         |
| OQ-B1: $p = 2$                                    | CLOSED — ZP-B T0                                                                                                                                                                                                                                                                    | Derived from AX-B1 and MP-1.                                                                                                                                                                                                                                                                                                                             |
| OQ-C1: Non-conservatism of DF                     | CLOSED — ZP-C T2                                                                                                                                                                                                                                                                    | Infinite sequence divergence proven. No postulates remain.                                                                                                                                                                                                                                                                                               |
| S1: Distribution stipulation                      | CLOSED — ZP-C T1                                                                                                                                                                                                                                                                    | Derived from AX-B1 and RP-1.                                                                                                                                                                                                                                                                                                                             |
| OQ-E1: Sequence vs. tree                          | CLOSED given the occurrence commitment — DA-2                                                                                                                                                                                                                                       | The structure is a forward-directed tree, not a linear sequence. Given that instantiation occurs — the occurrence commitment, not a consequence of T-SNAP — DA-2 supplies the succession and the branching follows. Countable vs. uncountable branching is perspective-dependent (DA-3).                                                                 |
| OQ-E2: Cardinality-semilattice correspondence     | OPEN                                                                                                                                                                                                                                                                                | Do specific semilattice structures correspond to specific cardinality regimes? Can the framework make predictions about which instantiations satisfy CH and which do not?                                                                                                                                                                                |

| Item                  | Status                                                                                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Remaining axioms      | INTENTIONAL — AX-B1, AX-G1, AX-G2 (named); A4, quine_unique, bot_self_mem, botCode_is_quine (typeclass fields) | Named commitments: AX-B1 (binary existence), AX-G1 (initial object), AX-G2 (source asymmetry). Structural commitments as typeclass fields: A4/bot_join (ZPSemilattice), AFAStructure.quine_unique and bot_self_mem (the Quine-atom role, which every ZP-A lattice supplies trivially), KleeneStructure.botCode_is_quine (computational closure). Typeclass fields are assumptions carried as hypotheses by the theorems that use them; #print axioms does not surface them. Further commitments are not axioms, among them CC-1 ( $S_0 = \perp$ ), a ZP-A Conditional Claim; T-SNAP's are stated under Premises of T-SNAP (DA-1 insert, § V). |
| Temperature T in BA-1 | PARAMETER — intentional                                                                                        | Universe-contingent. Physical predictions explicitly conditional on instantiation-specific T.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## Updated Traceability Register

| Claim                                   | Grounded In                                                                                                             | Bridge Axiom? | Status                                                                                                                                                                                                        |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Binary Snap causality                   | Shape: ZP-C D1, L-RUN, TQ-IH; ZP-A D2; T-SNAP premises: DA-1 insert § V. Occurrence: the occurrence commitment and DA-1 | None          | Derived for the shape — T-SNAP ✓; the snap occurs given the occurrence commitment and DA-1 (was: Axiomatic — AX-1)                                                                                            |
| DA-1: Instantiation = execution         | DP-2 (TrackedOutput, Snap.lean §VI); ZP-A CC-2, R3; ZP-C L-INF                                                          | None          | Derived Proposition — primary: DP-2 (da1_minimal_path proved axiom-free in Lean). Informal paths: CC-2/R3 (structural), L-INF (informational), AIT incompressibility (Path 3). AIT+ZF+AFA outside Lean scope. |
| DA-2: Instantiation succession          | ZP-A A4, CC-1; ZP-B C3, R1; T-SNAP                                                                                      | None          | Definitional Alignment — clarification of CC-1 scope; no new axiom                                                                                                                                            |
| C-DA2: Novelty of $\perp$ (conditional) | DA-2, ZP-B C3                                                                                                           | None          | CONDITIONAL — a modelling commitment. NOT derived: the separateness of the spaces is assumed, not obtained from C3, and snap_arc_z2_loop returns to the same 0.                                               |
| DA-3: Perspective-relative cardinality  | DA-2, C-DA2, ZP-B R1                                                                                                    | None          | Definitional (DA-3-D1, R-DA3-1); Candidate (DA-3-C1: outside-view inaccessibility)                                                                                                                            |

| Claim                               | Grounded In                                                                                                                                                                                                                                                                                                    | Bridge Axiom? | Status                                                                                                                                                                                           |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T-SNAP: Snap shape is derived       | L-RUN, TQ-IH (the T-BUF chain's lemmas); T-SNAP premises: DA-1 insert § V. DA-1 bears on occurrence, not on the shape                                                                                                                                                                                          | None          | Derived — Cross-Framework ✓                                                                                                                                                                      |
| AX-1 retirement                     | T-SNAP proves the shape; the snap occurs given the occurrence commitment and DA-1                                                                                                                                                                                                                              | N/A           | AX-1 is retired: T-SNAP proves the shape, and that the Snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2) |
| Iterative Forcing T5 (restated)     | Selection: succession_succ, ZeroParadox/Ordinal/SnapSuccession.lean § I; for a monotone map sending the tower's stages to $c_0$ , given the alignment hypothesis $h\epsilon_0$ , nothing below $\epsilon_0$ fires (snap_unconditional). Iteration: T3, R1, t_snap_irreversible, t_snap_holds_but_nothing_moves | None          | On an arbitrary lattice no step is selected. No step is guaranteed.                                                                                                                              |
| Multiverse — structural implication | T-SNAP + DA-2 jointly                                                                                                                                                                                                                                                                                          | None          | Structural implication given the occurrence commitment — T-SNAP gives the snap's shape, CC-2's undifferentiated $\perp$ the branching, DA-2 the succession                                       |
| OQ-E2: Cardinality correspondence   | DA-3                                                                                                                                                                                                                                                                                                           | N/A           | OPEN — formal derivation deferred                                                                                                                                                                |

## Validation Status

| Component                                             | Status / Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DA-1: Derived Proposition (Path 2 recharacterization) | Valid — DP-2 formal core: <code>da1_minimal_path</code> proved axiom-free in Lean (Snap.lean §VI). TrackedOutput separates output value from machine state; pre- and post-instantiation states are provably distinct even when both produce $\perp$ . ✓<br>ZP-K: <code>da1_closed_concrete : IsQuineAtom(<math>\perp</math> : MachinePhase)</code> proved in Kleene.lean. Path 1 is witnessed by <code>da1_closed_concrete</code> , which proves nothing computational; Path 3's witness is the <code>machinePhaseKleene</code> instance's <code>botCode_is_quine</code> field, a <code>KleeneStructure</code> requirement, not a second independent proof. <code>da1_paths_unified</code> carries the two together as a conjunction of witnesses; that they name one structural fact is the framework's reading. Path 2 (informational bridge, L-INF): FOUNDATIONAL COMMITMENT — a missing principle, not a missing proof. No computability library closes the gap between 'system at $P_0$ ' and 'system is running.' Forward paths: new axiom, Chalmers' implementation notion, or The Philosophical Question That Started This. ZP-M R-M.1 provides a retrospective structural analysis of why the gap resisted formalization. DA-1 does not depend on Path 2. CC-1 restated in ZP-J as an equivalence, and not forced (a valid sequence can start elsewhere on a carrier with a second point); CC-2 a Forced Metatheoretic Commitment whose Quine-atom role <code>t_exec_iff</code> shows only $\perp$ fills. |
| T-SNAP: Binary Snap derived                           | Valid — Derived, in two readings. The Lean form ( <code>t_snap_derived</code> ) fixes the SHAPE with no hypotheses and no axioms. The seven-step prose argument uses commitments, and that the transition OCCURS is not established by T-SNAP ( <code>tsnap_holds_but_nothing_moves</code> ). Premises: DA-1 insert § V, Premises of T-SNAP; two of them (CC-1 and occurrence at the first step) as hypotheses, <code>t_snap_given</code> . ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| AX-1 retirement                                       | Valid — AX-1 is retired, and its content was split in two. T-SNAP proves the shape, strengthened from assumed to derived. That the Snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2). Given CC-1, its first-step form is the hypothesis <code>hocc</code> in <code>t_snap_given</code> ; <code>tsnap_holds_but_nothing_moves</code> shows T-SNAP does not carry it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| DA-2: Instantiation Succession                        | Valid — Definitional Alignment. Clarification of CC-1 scope. No new axiom. A4 role of $\perp$ extended across instantiation boundaries. ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| C-DA2: Novelty of $\perp$ (conditional commitment)    | CONDITIONAL — a modelling commitment. NOT derived: the separateness of the spaces is assumed, not obtained from C3, which quantifies only over paths WITHIN one space. <code>snap_arc_z2_loop</code> measures against it — the 2-adic arc returns to the same 0. Must not be cited as derived.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Directed instantiation tree                           | Valid given the occurrence commitment — structural consequence of T-SNAP + DA-2. Once instantiation is taken to occur, branching follows and is not optional. Forward edges only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Branching tree structure                              | Valid given the occurrence commitment — T-SNAP + DA-2 jointly. T-SNAP fixes the snap's SHAPE and does NOT establish that it occurs; DA-2 establishes the branching tree structure. Universality (all outbound vectors) is DA-2's scope, not T-SNAP alone.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Component                                                                                                | Status / Notes                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monotonicity and path irrecoverability                                                                   | Valid — Structural consequence. Monotonicity (T3) constrains direction; additive ontology (R1) prohibits reduction. Path choice is undetermined by algebra.                                                                                                                                                |
| Ordinal direction of state sequences                                                                     | Valid — Derived from ZP-A T3 (monotonicity) and ZP-B C3 (irreversibility). Not assumed.                                                                                                                                                                                                                    |
| DA-3: Perspective-Relative Cardinality                                                                   | Valid (definitional components: DA-3-D1, R-DA3-1). Candidate (DA-3-C1: connection to specific set-theoretic independence results). OQ-E2 open.                                                                                                                                                             |
| DA-3-C1 — outside-view inaccessibility                                                                   | Candidate — no position within any instantiation can replicate the meta-level view of the branching structure. Formal derivation deferred to OQ-E2.                                                                                                                                                        |
| Remaining axioms: AX-B1, AX-G1, AX-G2; typeclass fields A4, quine_unique, bot_self_mem, botCode_is_quine | Named: AX-B1, AX-G1, AX-G2. Typeclass: A4/bot_join (ZPSemilattice), quine_unique + bot_self_mem (AFAStructure), botCode_is_quine (KleeneStructure). Assumptions carried as hypotheses by the theorems that use them; not surfaced by #print axioms. CC-1 is a further commitment (ZP-A Conditional Claim). |
| ZP-E results T1–T4, T6, T7                                                                               | Stated in an earlier version of this document (see the repository history) and not restated here. T5 is restated in DA-1 insert § VI.                                                                                                                                                                      |

ZP-E carries three formal inserts (DA-1, DA-2, DA-3). T-SNAP is derived, with its premises stated in the DA-1 insert § V and two of them (CC-1 and occurrence at the first step) as the hypotheses of `t_snap_given`; its irreversibility rests on ZP-A R1 and ZP-B C3, with ZP-G downstream. DA-1 is closed given DP-2 (`da1_minimal_path`, axiom-free). Path 1 is witnessed by `da1_closed_concrete` (ZP-K), which proves `IsQuineAtom( $\perp$  : MachinePhase)` and nothing computational; Path 3's witness is the `machinePhaseKleene` instance's `botCode_is_quine` field, a `KleeneStructure` requirement, not a second independent proof; and Path 2 is a foundational commitment (a missing principle, not a missing proof). Remark R- $\epsilon_0$  justifies the  $\epsilon_0$  symbol as a structural analogy. Remark R-AFA rules Foundation out by the self-membership of  $\perp = \{\perp\}$  (Regularity, `no_quine_atom`): the forcing and realizability of the AFA requirements are machine-checked (QuineHost), and a narrow Forced Metatheoretic Commitment remains. Open question: OQ-E2. Remaining axioms: AX-B1, AX-G1, AX-G2.